Benchmarking Federated Learning & Knowledge Distillation for Point Cloud Classification

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Background & Motivation

3D point cloud models are moving into clinics and onto robots, where two constraints collide: raw scans cannot leave the site that recorded them, and the deployed network must fit edge hardware. Federated learning (FL) [1] trains without pooling the data; knowledge distillation (KD) [2] then compresses the result for deployment. Yet non-IID client data degrades federated aggregation — documented for images, largely uncharted for hierarchical 3D feature extractors.

In practice the two stages are welded into one pipeline and judged end to end by the student's accuracy alone. **No systematic benchmark covers FL × KD for 3D point clouds** — and nobody asks whether that end-to-end number reflects the federated teacher at all.

Research question. When a federated 3D model is distilled for the edge, does the reported student accuracy measure the federated teacher — or something else?

504	−74.51%	+84.4
training runs, three seeds	student size vs. teacher	points above a collapsed teacher

Contributions

1. **First systematic FL × KD benchmark for 3D point clouds**: 13 FL algorithms × 10 KD objectives — the full 130-pair cross-product — at three seeds, **504 runs**, open-source.
2. Under extreme label skew the best FL teacher trails centralized training by **15.94–24.17 points**; all four server-side optimizers collapse to near-chance accuracy.
3. Distillation compresses the teacher into a student **74.51% smaller** and roughly **2× faster**; five of seven objectives match or beat the 92.44% teacher.
4. **Hard-label distillation masks federated failure**: a collapsed 8.50% teacher still yields a 92.94% student; label-free objectives track the teacher ($r \approx 0.99$).

Benchmark & Setup

Two deliberately contrasting testbeds: ModelNet40 [4] and a clinical craniosynostosis set [5] of patient head shapes (healthy control + coronal, metopic, sagittal suture fusions).

ModelNet40	40 classes; 9,843 / 2,468 shapes
Clinical	4 classes; 320 / 80 (stratified 80/20)
Input	1,024 points × 3 coords, no normals
Clients	5, disjoint label-skew slices
FL budget	20 rounds × 5 local epochs
Local opt.	Adam, lr 0.001, batch 24
KD stage	$\alpha=\beta=0.5$, $T=2.0$; 200 epochs
Metrics	instance acc.; mean class acc.
Seeds	$3 \times \{7, 42, 123\}$, mean±std

Best checkpoint per run; the label-skew partition is deterministic and fixed across seeds, so variance isolates training-time randomness. Communication is identical across strategies: 5.65 MB per transmission, 56.5 MB per round, ≈1.13 GB over 20 rounds. The data loader is dataset-agnostic — ModelNet10, OmniObject3D, YCB, and GazeboSim already run under the same interface.

Limitations & Future Work

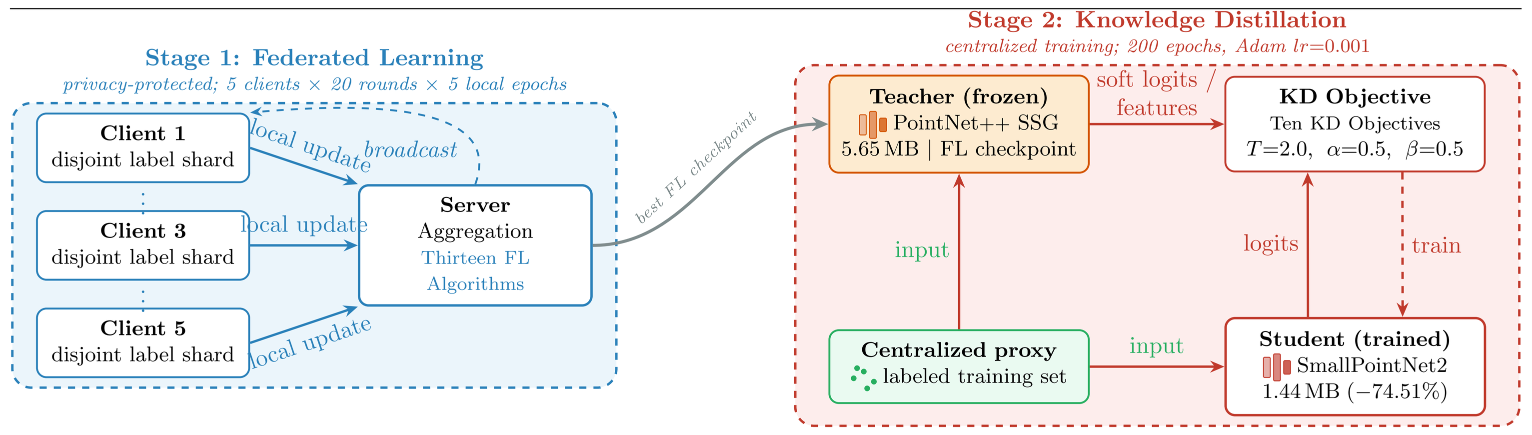
Limitations. Two datasets, one synthesized from a statistical shape model — larger scanned collections remain untested. One severe non-IID regime (disjoint label slices); softer partitions would narrow the gap. One backbone family (PointNet++ SSG); five clients, twenty rounds, homogeneous client models throughout.

Next steps: (i) a heterogeneity sweep over milder partitions; (ii) transformer and kernel-point backbones under federation; (iii) heterogeneous clients and larger real-world clinical scans.

References

[1] McMahan et al. Communication-Efficient Learning of Deep Networks from Decentralized Data. AISTATS, 2017.
[2] Hinton, Vinyals, Dean. Distilling the Knowledge in a Neural Network. arXiv:1503.02531, 2015.
[3] Qi et al. PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space. NeurIPS, 2017.
[4] Wu et al. 3D ShapeNets: A Deep Representation for Volumetric Shapes. CVPR, 2015.
[5] Schaufelberger et al. A Statistical Shape Model of Craniosynostosis Patients [...]. Zenodo, 2021.
[6] Lin et al. Ensemble Distillation for Robust Model Fusion in Federated Learning. NeurIPS, 2020.

Method Overview



Stage 1 trains a PointNet++ SSG [3] teacher across five privacy-protected clients; Stage 2 distills its best checkpoint into a compact student on a centralized labeled proxy — the objective's hard-label weight is the design choice under test.

How It Works

Two stages, one shared interface:

- **Stage 1 — federate:** five clients train on disjoint label slices; the server aggregates with one of **13 FL algorithms** (20 rounds).
- **Stage 2 — distill:** the best teacher checkpoint supervises a compact student on a labeled proxy split (**10 KD objectives**).
- Objectives split by hard-label weight: **six keep a CE term** ($w_{\mathrm{CE}}=0.5\text{–}1.0$); **four are pure transfer** ($w_{\mathrm{CE}}=0$: Feature, Attention, RKD, SP).
- Every cell re-trained per seed; the full 13 × 10 grid runs at multi-seed scale on the clinical task.

Teacher 5.65 MB → student 1.44 MB (−74.51%, ≈2× faster inference).

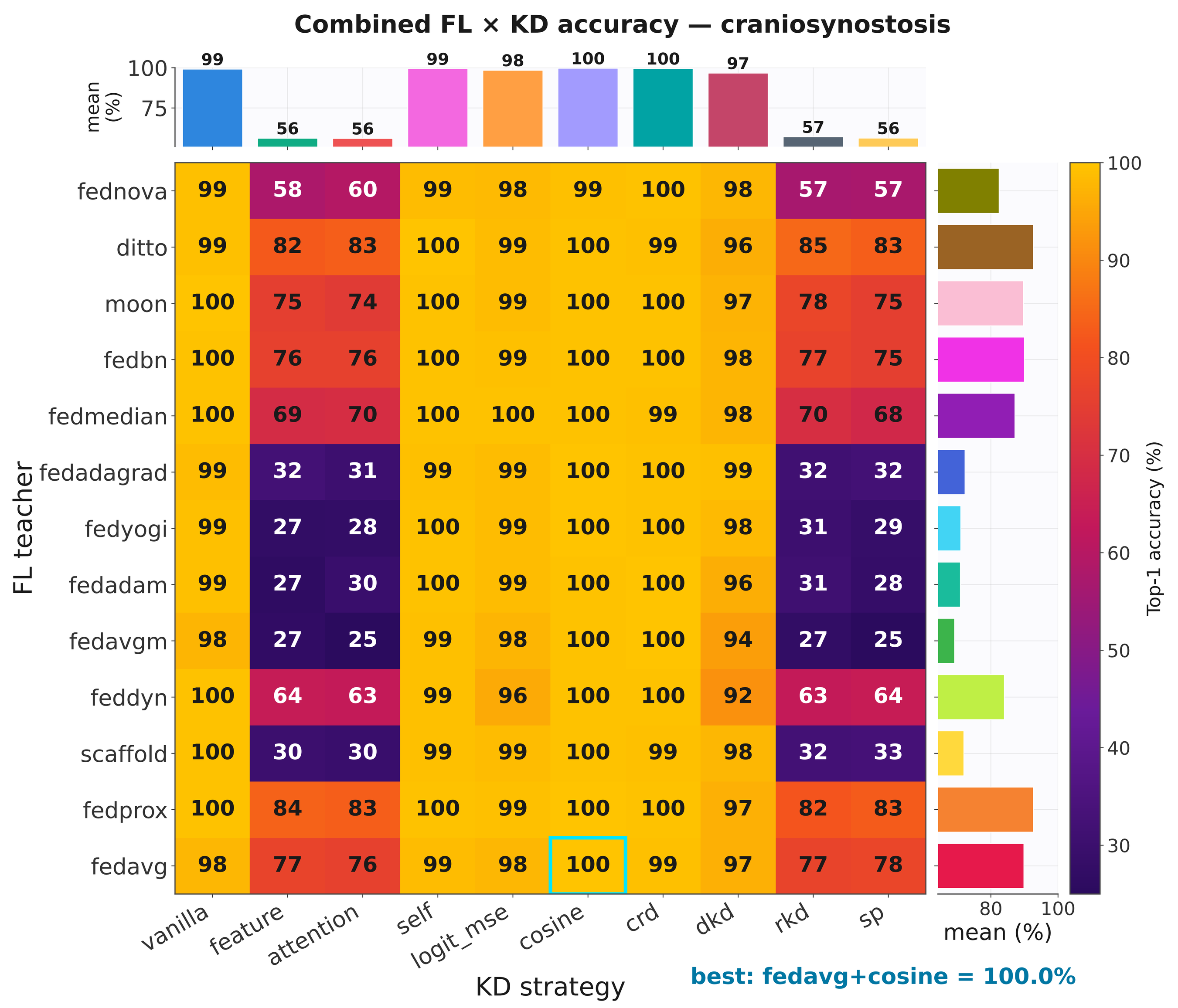
Why Hard Labels Deceive

Every objective optimises hard-label supervision plus teacher transfer on the proxy split:

$$\mathcal{L} = w_{\mathrm{CE}}\,\mathcal{L}_{\mathrm{CE}}\big(\hat{y},\, y\big) + \mathcal{L}_{\mathrm{transfer}}\big(f_S,\, f_T\big)$$

$\hat{y}$: student prediction; y: proxy label; f_S, f_T: student / teacher outputs or features. With $w_{\mathrm{CE}}>0$ a collapsed teacher contributes little gradient, the well-conditioned CE term dominates, and the student simply learns the proxy labels. At $w_{\mathrm{CE}}=0$ the teacher is the only signal — an honest, label-free probe of teacher quality.

The Full Clinical Grid, Multi-Seed



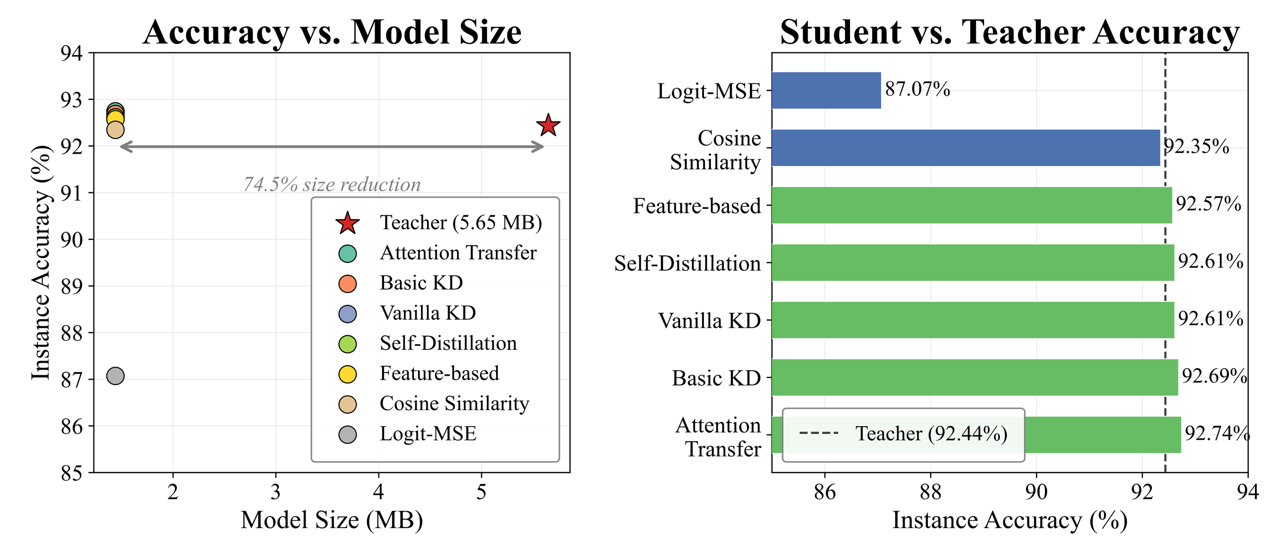
All 130 teacher–objective pairs [5]: the four $w_{\mathrm{CE}}=0$ columns (feature, attention, RKD, SP) track the teacher ($r=0.986\text{–}0.990$); the six hard-label columns stay at roughly 92–100% for every teacher — FedAvgM, frozen at the 25% chance level, still yields a 97.5% student under Vanilla KD and 100% under CRD.

Stage 1 Alone: FL Degrades Sharply

Method	ModelNet40 ↑	Craniosyn. ↑
Centralized (ref.)	$92.26_{\pm 0.04}$	$100.00_{\pm 0.00}$
FedNova	$\mathbf{76.32}_{\pm 1.55}$	$50.83_{\pm 8.04}$
FedProx	$71.04_{\pm 0.87}$	$\mathbf{75.83}_{\pm 10.63}$
FedDyn	$66.00_{\pm 1.28}$	$58.33_{\pm 8.13}$
SCAFFOLD	$64.26_{\pm 3.75}$	$36.67_{\pm 1.44}$
FedAvg [1]	$58.51_{\pm 0.25}$	$69.58_{\pm 2.60}$
FedAvgM	$4.05_{\pm 0.00}$	$25.00_{\pm 0.00}$

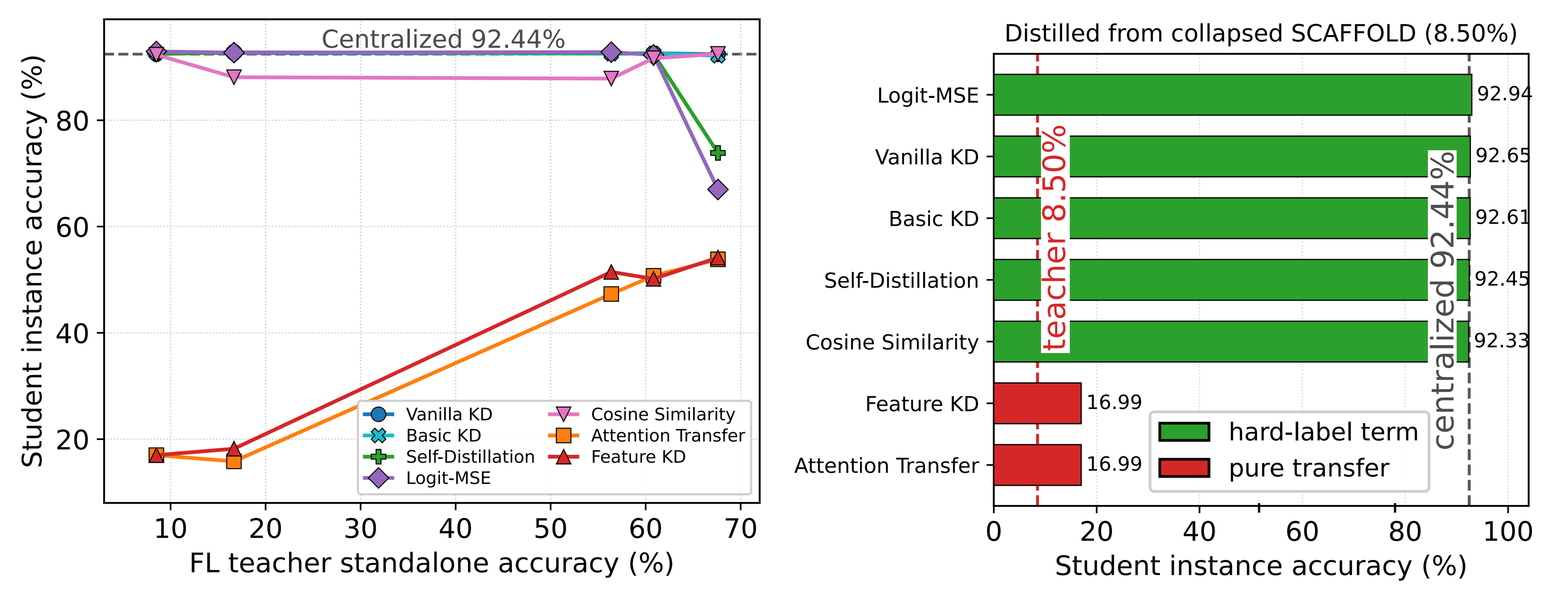
Instance accuracy (%, best per run, mean±std over three seeds). The best FL method trails centralized by 15.94 and 24.17 points, the ranking flips across datasets, and the four server-side optimizers collapse to 4.05–6.28% / 25.00–26.25% — near chance.

Stage 2 Alone: KD Compresses Cleanly



Distilled from the centralized 92.44% ModelNet40 teacher: every student is 74.51% smaller; five of seven objectives match or exceed the teacher, Cosine trails by 0.09 points, and the lone outlier is Logit-MSE (87.07%).

Combined: The Recovery Illusion



ModelNet40 focused round (single checkpoints): hard-label objectives stay at ≈92% for every teacher; $w_{\mathrm{CE}}=0$ objectives fall in step with their teachers.

*A SCAFFOLD teacher collapsed to 8.50% still gives a **92.94%** student — +84.4 points above its teacher.*

Same dichotomy at multi-seed scale on the clinical grid: CE-retaining objectives vary by at most 7.5 points across all thirteen teachers (r from −0.53 to +0.37); pure-transfer objectives span 57.5–58.3 points and track the teacher at r=0.986–0.990.

Take-away

Hard labels mask federated failure. Any objective with a CE term lifts the student to the centralized ceiling from *any* teacher — 8.50% → **92.94%** on ModelNet40, chance-level clinical teachers → 97.5–100% students — while reusing the very labels federation was meant to protect. The reported number measures the proxy labels, not the federated model.

What to do. Evaluate FL–KD pipelines **label-free** [6]: drop the CE term or distill on an unlabeled proxy, so that reported accuracy reflects the federated teacher ($r \approx 0.99$) rather than the proxy.